HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** microservices for enterprise applications, implementing RESTful APIs to support core business functions.
- Engineered data pipelines using **Python** and **SQL** to automate ETL processes, reducing manual data extraction and transformation efforts by approximately **40%**.
- Managed production incidents for critical enterprise systems, performing root cause analysis and implementing fixes within strict **SLA** targets, resolving over **150** critical issues.
- Designed and implemented constraint-based models of human metabolism for precision medicine applications, integrating biological and experimental data to understand disease pathways.
- Constructed interactive **Power BI** dashboards and reports, translating complex metabolic and nutritional data into actionable insights for research stakeholders.
- Applied statistical analysis and hypothesis testing using **R** to interpret experimental results and validate computational model predictions.
- Containerized computational modelling services using **Docker** and deployed them to cloud infrastructure, ensuring consistent environments for research reproducibility.

SKILLS

Modeling & Simulation – Constraint-based modelling, Metabolic modelling, Computational modelling, Digital Twin

Programming Languages – Python, Java, SQL, R

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Pandas, NumPy, Statistical analysis, Data visualization

Backend Development – Java Spring Boot, REST APIs, Microservices, Docker

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

Metabolic Pathway Modelling for Precision Nutrition

- Developed Python scripts to process and clean multi-omic datasets (genomics, metabolomics) for input into constraint-based models.
- Built and validated computational models of human metabolic networks, integrating dietary intake data to predict personalized nutritional responses.
- Visualized model outputs and simulation results using custom scripts and libraries, facilitating interpretation for biological researchers.

Personalised Health Insight Dashboard

- Designed and implemented a **Power BI** dashboard to present key metabolic indicators and personalized health recommendations derived from patient data.
- Developed complex **DAX** measures and data models to aggregate and analyze longitudinal health metrics for individual users.
- Collaborated with research leads to define reporting requirements and ensure the dashboard accurately reflected computational model insights.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)